

SFWRENG-3MX3 Assignment 5

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1) Impulse to Frequency Response

Use the Discrete Time Fourier Transform (DTFT) or the Continuous Time Fourier Series (CTFT) to find the frequency response that corresponds to the following impulse responses:

1.a) $h(t) = \delta(t - 1) + 2\delta(t - 2)$

Solution:

This problem is simply a plug-and-chug, noting the important property that the Dirac-Delta function acts as an evaluator when under the integral and that $X(\omega)$ corresponds to the frequency response of a system.

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} (x(t)e^{-i\omega t}) dt \\ &= \int_{-\infty}^{\infty} (\delta(t - 1) + 2\delta(t - 2))e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} (\delta(t - 1)e^{-i\omega t} + 2\delta(t - 2)e^{-i\omega t}) dt \\ &= e^{-i\omega} + 2e^{-2i\omega} \end{aligned}$$

1.b) $h(t) = \begin{cases} \frac{1}{3}, & \text{if } n = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$

Solution:

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} (x(n)e^{-i\omega n}) \\ &= \frac{1}{3} + \frac{1}{3}e^{-i\omega} + \frac{1}{3}e^{-2i\omega} \end{aligned}$$

2) The Step Response

In control theory, people use the step response in addition to the impulse response. The reason for this is that the step response gives information on how a controlled system reacts to a change of its set-point, which is a desired state for the system to be in. The step function is given by:

$$\text{step}(n) = \begin{cases} 0, & \text{if } n < 0 \\ 1, & \text{otherwise} \end{cases}$$

and the step response is the output of an LTI system in its zero state when $x(n) = \text{step}(n)$.

2.a) How might we compute the step response from the impulse response?

Solution:

We know that the impulse response $h(n)$ is the output of an LTI system $y(n)$ when its input $x(n)$ is $\delta(n)$. Now, the step function looks like an infinite train of the Kronecker-Delta functions, meaning $\text{step}(n) = \sum_{k=0}^{\infty} (\delta(n-k))$. Now, consider the convolutional equation. We know that for any input $x(n)$, the output of an LTI system is expressible as:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} (h(n-k)x(k)) \\ &= \sum_{k=0}^{\infty} (h(n-k)) \\ &= \sum_{k=0}^n (h(n-k)) \end{aligned}$$

2.b) Compute the step response of the system given by $y(n) = x(n) + \frac{1}{2}y(n-1)$ *directly*.

Solution:

$$\begin{array}{lll} y(0) & = x(0) + \frac{1}{2}y(-1) & = 1 \\ y(1) & = x(1) + \frac{1}{2}y(0) & = \frac{3}{2} \\ y(2) & = x(2) + \frac{1}{2}y(1) & = \frac{7}{4} \\ y(3) & = x(3) + \frac{1}{2}y(2) & = \frac{15}{8} \\ \vdots & \vdots & \vdots \\ y(n) & = x(n) + \frac{1}{2}y(n-1) & = \frac{2^{n+1}-1}{2^n} \end{array}$$

Thus, the step response of the system is given by $y(n) = \frac{2^{n+1}-1}{2^n}$

2.c) Compute the step response of the system given by $y(n) = x(n) + \frac{1}{2}y(n-1)$ using the formula derived in **a)**.

Solution:

Using the formula we derived earlier, we can compute the step response from the system's impulse response as follows:

(I) Find the impulse response $h(n)$ of the system $y(n) = x(n) + \frac{1}{2}y(n-1)$.

Substituting in $x(n) = \delta(n)$, to find $h(n)$, we get that:

$$\begin{aligned} h(0) &= \delta(0) + \frac{1}{2}y(-1) &= 1 \\ h(1) &= \delta(1) + \frac{1}{2}y(0) &= \frac{1}{2} \\ h(2) &= \delta(2) + \frac{1}{2}y(1) &= \frac{1}{4} \\ h(3) &= \delta(3) + \frac{1}{2}y(2) &= \frac{1}{8} \\ \vdots & & \vdots \\ h(n) &= x(n) + \frac{1}{2}y(n-1) &= \frac{1}{2^n} \end{aligned}$$

(II) Find the step response of the system using the impulse response $h(n)$.

Using both the formula derived in **a)** and the impulse response found in **(I)**, we find that the step response of the system is given by:

$$\begin{aligned} y(n) &= \sum_{k=0}^n (h(n-k)) \\ &= \sum_{k=0}^n \left(\frac{1}{2^{n-k}} \right) \\ &= \sum_{k=0}^n \left(\frac{2^k}{2^n} \right) \\ &= \frac{1}{2^n} \sum_{k=0}^n (2^k) \\ &= \frac{1}{2^n} \frac{2^{n+1} - 1}{2 - 1} \\ &= \frac{2^{n+1} - 1}{2^n} \end{aligned}$$

This corresponds with the step response found in **b.)**.

3) Filter Design

Design a simple FIR filter that removes any frequency component at $\frac{\pi}{2}$ and passes any constant signal with a gain of 1. Give the difference equation and state space representations for such a filter.

Solution:

4) Filtering Signals

What is the the output of a system with the frequency response:

$$H(\omega) = \begin{cases} 1, & \text{if } |\omega| \leq 2.5 \text{ radians/second} \end{cases}$$

when its input is a sawtooth signal, the periodic extension of the signal:

$$x(t) = \begin{cases} t, & \text{if } 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Hint: The sawtooth signal is 1-periodic, hence we can use the Fourier series to determine its frequency content.

Solution: